Supplementary Material S2

Basis or map? A pre-registered decomposition of the error of reduced-order and neural-operator surrogates of non-isothermal adsorption columns on unseen adsorbents

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This supplement holds supporting detail for the main text: each section expands a statement the main text summarises, with the same numbers. Section, figure, table and equation numbers without an "S" refer to the main text. Supplementary S1 is the correction ledger, whose entry codes (An, Bn) are cited below.

The terms are those of the main text (Section 1 and its glossary). The *legacy design* is the first dataset, with 12 held-out materials; the *larger design* is the second, 240 materials in five folds (Section 5). The rungs L1–L7 are the seven pre-registered candidate explanations of the error (Section 7). The *anchor* is the comparison of the reference solver with a measured MOF-303 bed (Section 4.3). The *edge rule* says that a tuning optimum found at the edge of its grid does not count until the grid is extended past it.

S1 Henry's law across the parameter space

S1.1 Henry's law holds everywhere; the region where it dominates does not

Across all 240 sampled materials, $n > 1$ strictly for every material, so the cooperative term is $o(c)$ at the origin and the Henry constant is finite and positive everywhere. But the Henry constant $K_H = q_{\max} f_H b_H$, the slope of equation (6) at the origin, spans 0.0322–2.52 $\mathrm{mol\,kg^{-1}}$ per $\mathrm{mol\,m^{-3}}$, 1.9 decades, median 0.359. The concentration at which the cooperative term reaches 1 % of loading (the top of the accessible Henry region) reaches $10^{-0.9}$ of the feed at best and a median of $10^{-2.0}$, but lies below 10^{-3} of the feed for 73 materials and below 10^{-6} for 34. For those the isotherm is effectively Sips-like at every concentration the column visits. Since concentration is proportional to relative humidity at fixed temperature, the near-linear uptake regime of a median material ends at $10^{-2.0}$ of the feed humidity. The design guarantees thermodynamic validity everywhere, not an experimentally accessible linear regime.

S2 Where real water-harvesting frameworks sit

The sampled space is meant to contain the adsorbents a water harvester would use, so we placed published ones in it (Figure S1), on the four axes that are properties of the framework rather than of the pellet and the packing: step position, capacity, heat of adsorption and the fraction of capacity taken below the step. The values come from the primary literature, MOF-303 [1, 2], MOF-801 [3], CAU-10-H [4, 5], aluminium fumarate [1, 6], MIL-160(Al) [7–9] and Co_2Cl_2(BTDD) [10]; each point spans the values its sources report, which differ in sample, temperature and in how the step is defined, and the below-step fractions are mostly read from published figures and are rough. Of the 6 frameworks, 3 lie inside the sampled ranges on every axis. The exceptions are at the edges, and one is far outside: MOF-801's step, at a

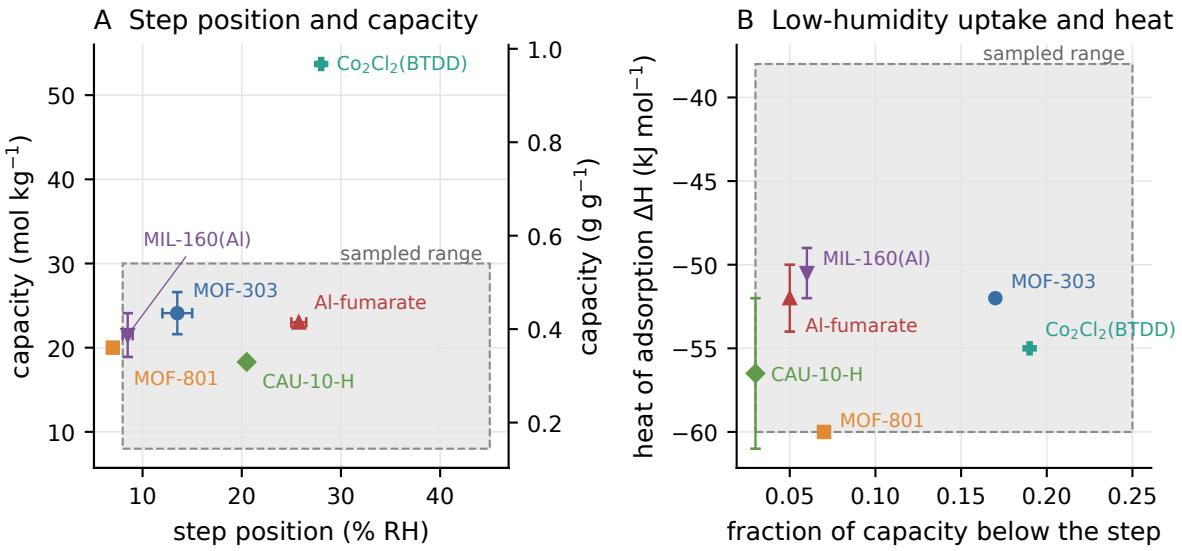


Figure S1: Published water-harvesting frameworks against the sampled parameter space (shaded boxes). **A** Step position and capacity. **B** Fraction of capacity taken below the step and heat of adsorption ΔH (negative: adsorption is exothermic). Bars span the values reported across sources; values, locations in each paper and confidence are recorded in `research/water_mof_parameters.json`.

relative humidity of about 0.07, sits just below the sampled floor of 0.08; one reported heat for CAU-10-H, $-61\,\mathrm{kJ\,mol^{-1}}$, lies just beyond the sampled range; and Co_2Cl_2(BTDD), at about $54\,\mathrm{mol\,kg^{-1}}$, holds 1.8× the largest capacity we sampled, 30. The verdicts of this paper are therefore statements about frameworks like MOF-303, aluminium fumarate and MIL-160, and should not be carried to record-capacity materials without new data. The cooperativity exponent n, the steepness of the step and arguably the property a harvester designer cares about most, is not placed: the sources report isotherms, not fitted step exponents, and fitting equation (6) to digitised curves would be a new measurement, not a reading of the literature.

S3 The tuning envelope of rung L3

The tuning envelope is itself a result

The families optimise 3.0 orders of magnitude apart: the Chebyshev family at 0.03 and the perceptron at 3×10^{-5}. 3 configurations failed on every seed and 4 more on some seeds; all are excluded and listed, never averaged in. Every family, the perceptron included, was tuned over the same learning-rate grid; none was left at a default.

S4 Basis size on the legacy design

The legacy result below is retracted as a verdict (A29) and kept as a measurement. At matched parameter counts and each family at its own best learning rate over 8 rates spanning 2.8 decades, DeepONet looked flat in basis size (Figure 6): -0.00102 from $p = 8$ to $p = 128$, interval $[-0.00585, 0.00491]$, which spans zero. Stated as what it excludes: a sixteenfold increase in basis size could not have improved this arm by more than 19 % of its error. The larger design found an improvement well inside that bound, which is exactly why a null on 12 held-out materials could not be the last word. Meanwhile its own POD floor falls from 1.44×10^{-2} to 5.02×10^{-4}, a factor of 28.7. The best DeepONet found anywhere, at the smallest basis, sits 53×

above the floor of the largest. The edge rule is the rule this rung was most exposed on: at the two smallest bases DeepONet's best learning rate first sat on the top of the grid, still moving. The grid was extended one rate upward for those arms, the optimum stayed where it was and is now interior, and every selected rate in this rung is bracketed, so the Fourier operator of L5 is not compared with an under-tuned DeepONet. *That DeepONet is as accurate as an oracle handed the true coefficients of its first 4 modes.*

S5 The scope of the L6 null

L6 predicted that the benefit of a separate kinetic object grows as the Damköhler number falls, and the main text finds no support for it. Kinetic control means Damköhler well below unity, and this dataset never goes there: its lowest per-material value is 7.7. The hypothesis predicted that separating the kinetic object pays most where kinetics dominate, and we have found no support for it over a range whose slow-kinetics end is merely *less fast*, not slow. A design that sampled Da < 1 could in principle recover the interaction. We did not sample it, and the reason is structural rather than an oversight: the kinetic coefficient is not a free parameter in this dataset but is derived from particle size through the Glueckauf relation, so Da $\propto d_p^{-2}$ and reaching Da $= 1$ from our slowest bed would take a pellet of about 9.5 mm. That is 11 pellet diameters along the 10 cm column, where a one-dimensional plug-flow model is no longer the right description. This is the same objection Section 12 raises against our own anchor bed. Decoupling k_{LDF} from d_p to reach that regime would restore precisely the physical inconsistency this dataset was built to remove. The null therefore holds across the range a physically consistent bed of this kind can occupy, and is not a statement about the kinetically-controlled limit.

S6 The power of the L6 test

Why the test has power now, and its limit

The between-material standard deviation of the arm difference fell to 0.0104, or 18.5 % of the base error, from 69 % previously, a 3.8× noise reduction from Sobol sampling and physically consistent kinetics. With 240 clusters instead of 12, that is what turned an unanswerable question into an answerable one. Power is 82 % at a 4 % effect and 26 % at 2 %, with size at the null 1.8 %; the minimum detectable effect at 80 % power is 4.0 %. The observed effect sits just under it, so the point estimate may be optimistic and the lower bound is the conservative reading.

S7 The mode count and the n-width

The mode count is scoped to a sample size

An earlier version of the paper stated the mode count as a property of the map. It is not: it is what a given number of training materials buys. Measured against material count (Figure 7), modes clearing $R^2 > 0.2$ grow from 2 to 13 over 12 to 192 materials and are still climbing. Three counts of modes clearing $R^2 > 0.2$ appear in this paper, and they are the same statistic under different conditions: 9 on the legacy design, and 13 and 17 on the larger one at the same number of training materials, the first for the learning-curve learner on its own basis and the second for the decomposition's regressor on the largest basis. They agree on what matters here: a learner predicts a modest set of mostly leading modes and little beyond it. The retraction is A21, and it withdrew the replacement text for an earlier retraction: the same inference, made twice.

The n-width is real and was not what binds

The residual *energy* of this snapshot family decays algebraically, as $k^{-2.45}$ in the number of modes k, with $R^2 = 0.9910$ over $k = 8$–64. The Kolmogorov n-width, whose n is this

mode count, is the residual *norm*, the square root of that, so it decays as $k^{-1.23}$ and halving the error requires 1.76× the modes. We separate the two because an earlier draft quoted the energy exponent and called it the n-width, next to a halving factor computed correctly from the norm: two numbers in one sentence that disagreed by a factor of two in the exponent. Any method whose output is a linear combination of a fixed basis is bounded below by it. Our frozen protocol called this the central result of the ladder and predicted that DeepONet would track it. The prediction was wrong: the bound is real, and it is inactive. Withdrawing it is A17. For transport-dominated problems the slow decay is established: Ohlberger and Rave [11] prove a *lower* bound of $\frac{1}{2}k^{-1/2}$ for linear advection with jump discontinuities, which is a bound and not a rate. Our measured decay is faster than that bound, and the reason is the hypothesis: their fields carry jump discontinuities, ours are smoothed by axial dispersion and by finite-rate kinetics, so the worst case their theorem constructs does not occur here. That is also why the bound can be simultaneously real and inactive; this paragraph has to hold both statements together.

S8 Monotonicity of the predicted fronts

The two-front frame of Section 9 predicts two front trajectories per sample. The obvious first attack on its result is that the front predictor does not know the constraints the true trajectories satisfy exactly. It does violate them, in 55 % of samples at the lower landmark, while the true trajectories violate them in none. But projecting onto the constraint set changes R^2 by at most 1.9×10^{-4} and makes the reconstruction significantly *worse*: 0.04770 against 0.04812, a difference of −0.00041 with interval [−0.00070, −0.00010] (tighter than Table 4's because the projected arm is a deterministic transformation of the raw arm, so the between-material variance cancels). The violations are frequent but negligible; the remaining headroom is the front predictor's raw accuracy.

S9 The dispersion closure, and the limits of the isotherm and the parameter space

Our datasets use $D_L = 0.7D_m + 0.5d_pu$, with $u = v/\varepsilon_t$ the interstitial velocity and D_m the molecular diffusivity of water in air, a form due to Wakao and Funazkri [12] that Perry's handbook [13] presents as a *lower bound* for *nonporous* particles; for porous, strongly adsorbing particles such as ours the effective axial dispersion is larger. The form is also the high-particle-Péclet *limit* of Edwards and Richardson's correlation [14], $D_L = 0.73D_m + 0.5\,d_pu\,[1 + 9.7\,D_m/(d_pu)]^{-1}$, whose mechanical term carries the extra bracketed factor, so we measured whether the limit applies. The particle Péclet number d_pu/D_m runs 4.6–190 across the dataset, median 49, spanning 1.62 decades. Over that range the two published forms disagree by a median factor of 1.19, exceeding 1.1× for 86.3 % of samples and 1.5× for 9.9 %, with a worst case of 2.04. The disagreement is bounded and modest for the dataset, but it is not modest for the anchor.

The anchor bed sits at a particle Péclet number of 2.9, and the disagreement between the two forms is maximised at 3.8, where it reaches 2.06×. In other words the one bed in this paper that is compared against an experiment is also the one place where the truncation matters most: there our closure disperses 2.02× more than the correlation it is the limit of, taking the column Péclet number from 2.32 to 1.15. Re-solving that bed under the untruncated form (a labelled post-hoc sensitivity, not a fit, and not the pre-registered result) moves the 95 % arrival time from 431 to 384 minutes against 354 measured, the plateau from 0.336 to 0.293 against 0.267, and the error against the digitised effluent from 0.128 to 0.105 (Figure S2). The authors' own Bruggeman closure goes further, to 0.083, but overshoots in the other direction at 329 minutes; the measured value is bracketed by the two closures. A substantial part of the anchor's second

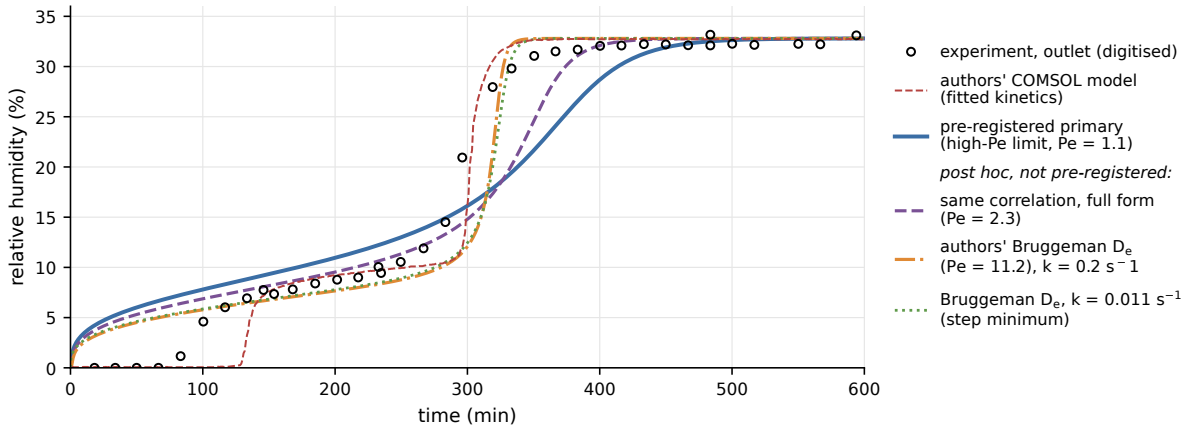


Figure S2: Post-hoc dispersion sensitivity on the anchor bed, labelled as such and reported as a limitation rather than as a result. The pre-registered primary uses the high-Péclet limit of the correlation; the second curve is the *same* correlation without that truncation, at the same particle size, velocity and molecular diffusivity; the third is the authors' own Bruggeman closure for the effective dispersion D_e; the fourth repeats that closure at the smallest LDF constant k (in s^{-1}) that still keeps the step. Pe is the column Péclet number. The measured 95 % arrival time is bracketed by the untruncated correlation and the authors' Bruggeman closure.

named discrepancy is therefore one term evaluated at its limit: the untruncated form removes 61 % of the model's excess 95 % arrival time and 18 % of its error against the digitised effluent (the error is dominated by the first-wave timing, which no dispersion closure touches). None of this affects a surrogate result, because every arm sees the same reference; it bounds how far the reference may be read as physical.

The isotherm's low-humidity branch is too weak

The Henry fraction was fitted to the step position and the capacity, never to dynamics; the anchor comparison shows the model holds too little water below the step.

Part of the heat range is occupied by no placed framework

The sampled heat of adsorption runs from -60 to $-38\,\mathrm{kJ\,mol^{-1}}$; the water-harvesting frameworks we placed (Figure S1) report -61 to -49. Near its least exothermic edge the sampled range gives an isosteric heat ($-\Delta H + RT$) below water's latent heat of vaporisation at room temperature, so that water is bound more weakly than in its own liquid. That is the regime of hydrophobic adsorbents rather than of the frameworks this paper is motivated by. The range was pre-registered and is kept; verdicts averaged over it weight that region as heavily as the occupied one.

Not every sample reaches the step

A harvester is operated above its step by design; the sampled space is not. 14 % of the accepted samples of the larger dataset (18 % of the legacy one) are fed below their material's step humidity, so the feed never reaches pore filling and the breakthrough is the primary-site front alone. These samples are in every verdict; a surrogate built only for harvesting conditions would not need them.

The step does not move between feed temperatures

The cluster affinity is set per run so that the step sits at the material's step humidity at the feed temperature (Section 2). Within a run the step still moves with the local bed temperature through the heat of adsorption, but across runs a material's step humidity at the feed temperature is the same by construction. In real MOF-303 it moves with temperature [15].

The datasets therefore do not test whether a surrogate transfers the temperature dependence of the step.

Three parameters are not measurements

Pellet density and bed porosity are packing values, not MOF-303 measurements [16]. No citable measured heat capacity for MOF-303 was found, so it is reported as a sensitivity band rather than as a cited constant.

No covalent organic framework case

We found no published breakthrough measurement on a covalent organic framework (COF) suitable for this comparison, so no claim is made for that class.

References

[1] N. Hanikel, M. S. Prévot, F. Fathieh, E. A. Kapustin, H. Lyu, H. Wang, N. J. Diercks, T. G. Glover, O. M. Yaghi, Rapid cycling and exceptional yield in a metal–organic framework water harvester, ACS Central Science 5 (10) (2019) 1699–1706. doi:10.1021/acscentsci.9b00745.

[2] F. Fathieh, M. J. Kalmutzki, E. A. Kapustin, P. J. Waller, J. Yang, O. M. Yaghi, Practical water production from desert air, Science Advances 4 (6) (2018) eaat3198. doi:10.1126/sciadv.aat3198.

[3] H. Furukawa, F. Gándara, Y.-B. Zhang, J. Jiang, W. L. Queen, M. R. Hudson, O. M. Yaghi, Water adsorption in porous metal–organic frameworks and related materials, Journal of the American Chemical Society 136 (11) (2014) 4369–4381. doi:10.1021/ja500330a.

[4] M. V. Solovyeva, A. I. Shkatulov, L. G. Gordeeva, E. A. Fedorova, T. A. Krieger, Y. I. Aristov, Water vapor adsorption on CAU-10-*X*: effect of functional groups on adsorption equilibrium and mechanisms, Langmuir 37 (2) (2021) 693–702. doi:10.1021/acs.langmuir.0c02729.

[5] D. Fröhlich, E. Pantatosaki, P. D. Kolokathis, K. Markey, H. Reinsch, M. Baumgartner, M. A. van der Veen, D. E. De Vos, N. Stock, G. K. Papadopoulos, S. K. Henninger, C. Janiak, Water adsorption behaviour of CAU-10-H: a thorough investigation of its structure–property relationships, Journal of Materials Chemistry A 4 (30) (2016) 11859–11869. doi:10.1039/c6ta01757f.

[6] E. Laurenz, G. Füldner, L. Schnabel, G. Schmitz, A novel approach for the determination of sorption equilibria and sorption enthalpy used for MOF aluminium fumarate with water, Energies 13 (11) (2020) 3003. doi:10.3390/en13113003.

[7] M. Solovyeva, I. Krivosheeva, L. Gordeeva, Y. Aristov, MIL-160 as an adsorbent for atmospheric water harvesting, Energies 14 (12) (2021) 3586. doi:10.3390/en14123586.

[8] T. Aumond, M. Bonneau, C. Daniel, M. Perbet, F. Meunier, D. Farrusseng, Adsorption performances of water-stable MIL-160(Al) at scale: an alternative to FAM-Z02 in water harvesting and chillers, ACS Omega 11 (10) (2026) 16294–16299. doi:10.1021/acsomega.5c11973.

[9] D. Lenzen, J. Zhao, S.-J. Ernst, M. Wahiduzzaman, A. K. Inge, D. Fröhlich, H. Xu, H.-J. Bart, C. Janiak, S. Henninger, G. Maurin, X. Zou, N. Stock, A metal–organic framework for efficient water-based ultra-low-temperature-driven cooling, Nature Communications 10 (1) (2019) 3025. doi:10.1038/s41467-019-10960-0.

[10] A. J. Rieth, S. Yang, E. N. Wang, M. Dincă, Record atmospheric fresh water capture and heat transfer with a material operating at the water uptake reversibility limit, ACS Central Science 3 (6) (2017) 668–672. doi:10.1021/acscentsci.7b00186.

[11] M. Ohlberger, S. Rave, Reduced basis methods: success, limitations and future challenges, in: Proceedings of the Conference Algoritmy 2016, 2016, pp. 1–12. arXiv:1511.02021.

[12] N. Wakao, T. Funazkri, Effect of fluid dispersion coefficients on particle-to-fluid mass transfer coefficients in packed beds, Chemical Engineering Science 33 (10) (1978) 1375–1384. doi:10.1016/0009-2509(78)85120-3.

[13] M. D. LeVan, G. Carta, J. A. Ritter, K. S. Walton, Adsorption and ion exchange, in: D. W. Green, M. Z. Southard (Eds.), Perry's Chemical Engineers' Handbook, 9th Edition, McGraw-Hill Education, 2019, Section 16.

[14] M. F. Edwards, J. F. Richardson, Gas dispersion in packed beds, Chemical Engineering Science 23 (2) (1968) 109–123. doi:10.1016/0009-2509(68)87056-3.

[15] J. Hastings, T. Lassitter, Z. Zheng, S. Chheda, J. I. Siepmann, L. Gagliardi, O. M. Yaghi, T. G. Glover, High-temperature water adsorption isotherms and ambient temperature water diffusion rates on water harvesting metal–organic frameworks, The Journal of Physical Chemistry C 128 (27) (2024) 11328–11339. doi:10.1021/acs.jpcc.4c01733.

[16] N. Hanikel, X. Pei, S. Chheda, H. Lyu, W. Jeong, J. Sauer, L. Gagliardi, O. M. Yaghi, Evolution of water structures in metal–organic frameworks for improved atmospheric water harvesting, Science 374 (6566) (2021) 454–459. doi:10.1126/science.abj0890.